MODULE 02 · WEB

JavaScript Fundamentals

A Custom AI-Generated Companion Guide

The language that makes websites interactive — from a button click to a full web app.

DIGITAL GIRLS INITIATIVE PAKISTAN
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8–10 weeks · 5–7 hrs/week

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Welcome & How to Use This Guide

JavaScript is what turns a static page into an app — it's the language behind every dropdown menu, form validation, and live-updating feed you've ever used. This track takes you from your first variable to fetching real data from the internet and building an interactive project of your own.

> **Mentor Tip**
> Break early and often — you'll learn more from figuring out why your code doesn't work than from code that works on the first try.

Estimated time: 8–10 weeks · 5–7 hrs/week · Level: HTML & CSS basics recommended

01 Syntax & Core Concepts

How JavaScript thinks

Every program is built from a small set of ideas repeated constantly: storing values, making decisions, and repeating actions.

01 Variables & data types
let, const, strings, numbers, booleans — how JavaScript stores information.

02 Operators & expressions
Arithmetic, comparison, and logical operators used to build any condition.

03 Conditionals
if / else / switch — how a program makes decisions.

04 Loops
for, while, and for...of — how a program repeats work without repeating code.

★ Practice Project
Build a console-based number guessing game using conditionals and a loop.

02 Functions & the DOM

Reusable logic and a live page

Functions let you package logic once and reuse it anywhere. The DOM is how JavaScript reaches into HTML and changes it live, in the browser.

01 Functions & scope
Declarations, parameters, return values, and arrow function syntax.

02 Selecting the DOM
querySelector and getElementById — grabbing the exact element you want.

03 Changing the DOM
Updating text, styles, and classes in response to code.

04 Event listeners
Responding to clicks, input, and form submissions in real time.

★ Practice Project

Build a click counter and a simple light/dark mode toggle button.

03 Data Structures & Logic

Working with real, structured data

Almost every app deals with lists and structured records — this phase covers the tools built for exactly that.

01 Arrays & array methods
map, filter, and reduce — the three methods used in nearly every JS codebase.

02 Objects
Key-value data, nested objects, and accessing properties.

03 JSON
The universal format used to move data between a server and your app.

04 Debugging with the console
console.log habits and reading browser error messages without panic.

★ Practice Project
Build a to-do list app: add, complete, and delete tasks, stored in an array and rendered to the DOM.

04 Async JavaScript & Projects

Talking to the real internet

Real apps fetch live data from servers. This phase covers the async patterns needed to do that cleanly.

01 Promises
Understanding pending / resolved / rejected states.

02 async / await
Writing asynchronous code that reads top-to-bottom, the modern standard.

03 The fetch API
Requesting real data from a public API.

04 Error handling
try / catch so a failed request doesn't crash your whole app.

★ Practice Project
Build a weather-lookup or quote-generator app that fetches live data from a free public API.

Free Video Courses

Curated, highly-rated, and completely free. Pick whichever teaching style fits you best.

> **freeCodeCamp — JS Algorithms & Data Structures**
> Full certification with built-in practice challenges
> *https://www.freecodecamp.org/learn/javascript-algorithms-and-data-structures/*

> **The Net Ninja — JavaScript Playlist**
> Beginner-paced, project-driven video series
> *https://www.youtube.com/@NetNinja*

> **javascript.info**
> The most thorough free written tutorial available
> *https://javascript.info/*

> **Traversy Media — JS Crash Course**
> Fast, practical crash courses and real mini-projects
> *https://www.youtube.com/@TraversyMedia*

Your Practice Checklist

Print this page or copy it into a notebook. Check off each project as you finish it — real, working projects are what turn this guide into a portfolio.

- [ ] **Phase 01: Syntax & Core Concepts — Build a console-based number guessing game using conditionals**
- [ ] **Phase 02: Functions & the DOM — Build a click counter and a simple light/dark mode toggle button.**
- [ ] **Phase 03: Data Structures & Logic — Build a to-do list app: add, complete, and delete tasks, stored in a**
- [ ] **Phase 04: Async JavaScript & Projects — Build a weather-lookup or quote-generator app that fetches li**

Finished all four phases?

Bring your projects to a DG·IP mentor session — we review portfolios and help you plan your next module.

You've got the roadmap.
Now go build something.

This guide is one part of the free DG·IP curriculum. Come back to the website for the live version of this module, curated videos, and seven other free skill tracks.

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